

2.3 Topic 3: The voice of the genome

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the practical and investigative skills identified in numbers 4 and 5 in the table of *How Science Works* on page 12 of this specification.
- 2 Distinguish between eukaryotic and prokaryotic cells in terms of their structure and ultrastructure.
- 3 Describe the ultrastructure of an animal (eukaryotic) cell (nucleus, nucleolus, ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes, and Golgi apparatus) and recognise these organelles from EM images.
- 4 Explain the role of the rough endoplasmic reticulum (rER) and the Golgi apparatus in protein transport within cells and including its role in formation of extracellular enzymes.
- 5 Describe how the cells of multicellular organisms can be organised into tissues, tissues into organs and organs into systems.
- 6 Explain the role of mitosis and the cell cycle for growth and asexual reproduction.
- 7 **Describe the stages of mitosis and how to prepare and stain a root tip squash in order to observe them practically.**
- 8 Explain the role of meiosis in the production of gametes and genetic variation through recombination of alleles and genes including independent assortment and crossing over (details of the stages of meiosis are not required).
- 9 Explain how mammalian gametes are specialised for their functions.
- 10 Describe the process of fertilisation in mammals and flowering plants (starting with the acrosome reaction in mammals and pollen tube growth in plants and ending with the fusion of the nuclei) and explain the importance of fertilisation in sexual reproduction.

- 11 Explain what is meant by the terms stem cell, pluripotency and totipotency and discuss the way society uses scientific knowledge to make decisions about the use of stem cells in medical therapies (e.g. regulatory authorities relating to human embryo research, ability of stem cells to develop into specialised tissues, potential sources of stem cells, who could benefit from the therapies, procedures to obtain stem cells and their risks).
- 12 **Describe how totipotency can be demonstrated practically using plant tissue culture techniques.**
- 13 Explain how cells become specialised through differential gene expression, producing active mRNA leading to synthesis of proteins, which in turn control cell processes or determine cell structure in animals and plants (details of transcription factors are not required at IAS).
- 14 Explain how a phenotype is the result of an interaction between genotype and the environment (e.g. animal hair colour, human height, monoamine oxidase A (MAOA) and cancers), but the data on the relative contributions of genes and environment is often difficult to interpret.
- 15 Explain how some phenotypes are affected by alleles at many loci (polygenic inheritance) as well as the environment (e.g. height) and how this can give rise to phenotypes that show continuous variation.